

Arif Ullah (Khan)

Ph.D. (Electro. and Commun. Engineering), MPEC, MIEEE

Tel: +44 (0) 28 9097 1700, Cell: +44 07****1**15

Email: [a\[dot\]ullah\[at\]qub\[dot\]ac\[dot\]uk](mailto:a[dot]ullah[at]qub[dot]ac[dot]uk), Gmail: arifkhaan.ciit

Office: Centre for Wireless Innovation (CWI) Queen's University Belfast Queen's Road, Belfast BT3 9DT UK.

Google Scholar Profile: <https://shorturl.at/lqzH1>



RESEARCH INTERESTS

- Holographic MIMO Communications; Electromagnetic Signal and Information Theory
- UAV and RIS Assisted Communications; Stochastic Geometry Analysis of Cellular Networks
- Machine learning and AI enabled Wireless Communications

EDUCATION

- GIK Institute of Engineering Sciences and Technology Topi, 23640, Pakistan**
Doctor of Philosophy (PhD) in Electronic Engineering Sep. 2017 - Jul. 2021
 - Study Emphasis: Wireless Communication and Networking
 - Dissertation: User-centric Small Cell Aided Cellular Networks: Sub-6GHz and Hybrid mmWave Communications
 - PhD Supervisor: [Prof. Dr. Ziaul Haq Abbas](#)
- COMSATS University Abbotabad, 22060, Pakistan**
Master of Science (MS) in Electrical Engineering Mar. 2014 - Jul. 2016
 - Study Emphasis: Wireless Communication and Signal Processing
 - Dissertation: Precise Estimation of Soft Output in MIMO OFDM Receiver using Modified LAS Algorithm
 - MS Supervisor: [Prof. Dr. Shahid Khattak](#)
- Balochistan University of IT, Engg. & Management Sci. Quetta, Pakistan**
Bachelor of Science (BS) in Electronic Engineering Aug. 2007 - Dec. 2011
 - Study Emphasis: Electronic and Communication Engineering
 - Final Year Project: Real time Tracking, Monitoring, and Controlling of Vehicles through GPS and GSM

PROFESSIONAL EXPERIENCE

- Queen's University Belfast BT7 1NN, United Kingdom.**
Research Fellow, Center for Wireless Innovations (CWI) Sep. 2024 - Present
- Chosun University Gwangju, 61452, Republic of Korea**
Research Assistant Professor Sep. 2023 - Aug. 2024
 - Department of Computer Engineering, College of IT Convergence:
- Chosun University Gwangju, 61452, Republic of Korea**
Assistant Professor, Wireless Communication and Networking Apr. 2022 - Aug. 2023
 - Department of Computer Engineering, College of IT Convergence:
- Chosun University Gwangju, 61452, Republic of Korea**
Postdoctoral Fellow, Smart Networking Research Laboratory (SNL) Oct. 2021 - Apr. 2022
- GIK Institute of Engineering Sciences and Technology Topi, 23640, Pakistan**
Graduate Teaching/Research Assistant Sep. 2017 - Jun. 2021
 - Teaching Assistant:** Assisted as a teaching assistant in the undergraduate courses such as Linear Circuit Analysis (EE-211) course (Fall 2017), Digital Control System (EE-444) course (Spring 2021)
 - Lab. Instructor:** Assisted as an instructor in the following undergraduate Labs: 1) Electronics Devices and Circuit Lab (EE-231L) (Spring 2018-2019), 2) Signal and System Lab (EE-351L) (Fall 2018-2019), 3) Communication System Lab (EE-361L) (Spring 2020)
- COMWAVE Institute of Information Sci. and Techno., Abbotabad, Pakistan**
Visiting Lecturer Jul. 2016 - Jul. 2017
- Egyptian Pakistani Telecommunication Services Company Limited Pakistan**
Trainee Engineer (Operation and Maintenance) Sep. 2012 - Jun. 2013

SELECTED PUBLICATION

Peer-reviewed Publications:

- [J16]: **Arif Ullah**, Fawad, Amir Nadeem, Muhammad Arif, Muhammad Mehran Bashir, and Wooyeol Choi, "6G Internet-of-Things Assisted Smart Homes and Buildings: Enabling Technologies, Opportunities and Challenges", *IEEE Internet of Things Journal*, vol. 32, p. 101658, 2025. DOI: [10.1016/J.IOT.2025.101658](https://doi.org/10.1016/J.IOT.2025.101658)
- [J16]: Pulok Tarafder, Chanjun Chun, **Arif Ullah**, Yonggang Kim, and Wooyeol Choi, "Channel Estimation in 5G-and-Beyond Wireless Communication: A Comprehensive Survey", *MDPI Electronics*, vol. 14, no. 4, p. 720, 2025. DOI: [10.3390/Electronics14040750](https://doi.org/10.3390/Electronics14040750)
- [J15]: Sufyan Ali, S. A. Ali Shah, K. B. Khan, Izaz Ali Shah, **Arif Ullah**, T.A. Siddiqui, Saiful Islam and Hyoungsuk Yoo, "Self-Decoupled 8-Element Dual Polarized MIMO Antenna for B5G Handsets With Sub-6 GHz and mmWave Integration," in *IEEE Access*, vol. 12, pp. 167324-167332, 2024, DOI: [10.1109/ACCESS.2024.3491867](https://doi.org/10.1109/ACCESS.2024.3491867).
- [J14]: Iftikhar Ahmad, **Arif Ullah**, and Wooyeol Choi, "WiFi-Based Human Sensing with Deep Learning: Recent Advances, Challenges, and Opportunities", *IEEE Open Journal of the Communication Society*, early access, 2024. DOI: [10.1109/OJCOMS.2024.3411529](https://doi.org/10.1109/OJCOMS.2024.3411529)
- [J13]: **Arif Ullah**, Wooyeol Choi, Teklu Merhawit Berhane, Yusuf Sambo, and Muhammad Ali Imran, "Soft-Output Deep-LAS Detection for Coded MIMO System: A Learning-Aided LLR Approximation", *IEEE Transactions on Vehicular Technology*, early access, 2024. DOI: [10.1109/TVT.2024.3391614](https://doi.org/10.1109/TVT.2024.3391614)
- [J12]: Aamir Nadeem **Arif Ullah**, and Wooyeol Choi, "Social-Aware Peer Selection for Energy Efficient D2D Communications in UAV-Assisted Networks: A Q-Learning Approach", *IEEE Wireless Communications Letter*, vol. 13, no. 5, pp. 1468-1472, 2024. DOI: [10.1109/LWC.2024.3375235](https://doi.org/10.1109/LWC.2024.3375235)
- [J11]: Youngwoo Oh, **Arif Ullah**, and Wooyeol Choi, "Multi-Objective Reinforcement Learning for Power Allocation in Massive MIMO Networks: A Solution to Spectral and Energy Trade-Off", *IEEE Access*, vol. 12, pp. 1172 - 1188, Dec. 2023. DOI: [10.1109/ACCESS.2023.3347788](https://doi.org/10.1109/ACCESS.2023.3347788)
- [J10]: **Arif Ullah**, Wooyeol Choi, and Sinem Coleri, "Path Loss Estimation and Jamming Detection in Heterogeneous Vehicular Networks: A Hybrid Machine Learning Framework", *IEEE Sensors Journal*, vol. 23, no. 24, pp. 31325 - 31336, 2023. DOI: [10.1109/JSEN.2023.3329490](https://doi.org/10.1109/JSEN.2023.3329490)
- [J9]: **Arif Ullah**, Wooyeol Choi, Ziaul Haq Abbass, and Ghulam Abbass, "Aerial-Terrestrial Networks with Multi-antenna Transmissions: How Many UAVs Need to Be Deployed?", *IEEE Transaction on Vehicular Technology*, vol. 73, no. 2, pp. 2212 - 2226, Sep. 2023. DOI: [10.1109/TVT.2023.3316195](https://doi.org/10.1109/TVT.2023.3316195)
- [J8]: Fawad, Iftikhar Ahmad, **Arif Ullah**, and Wooyeol Choi, "Machine Learning Framework for Precise Localization of Bleached Corals Using Bag-of-Hybrid-Visual-Feature Classification", *Nature Scientific Report*, vol. 13, pp. 1946, 2023. DOI: [10.1038/s41598-023-46971-7](https://doi.org/10.1038/s41598-023-46971-7)
- [J7]: **Arif Ullah**, Ziaul Haq Abbas, Ghulam Abbas, Fazal Muhammad and Jae-Mo Kang, "Hybrid millimeter wave heterogeneous networks with spatially correlated user equipments," *Digital Communications and Networks*, in press, Oct. 2022. DOI: [10.1016/j.dcan.2022.10.022](https://doi.org/10.1016/j.dcan.2022.10.022)
- [J6]: **Arif Ullah**, Ziaul Haq Abbas, Fazal Muhammad, Irfanullah, Alam Zeb, and Shahid Khattak, "Likelihood ascent search augmented sphere decoding receiver for MIMO systems using M-QAM constellations," *IET Communication*, vol. 14, no. 22, pp. 4152-4158, Dec. 2020. DOI: [10.1049/iet-com.2019.1316](https://doi.org/10.1049/iet-com.2019.1316)
- [J5]: **Arif Ullah**, Ziaul Haq Abbas, Ghulam Abbas, Fazal Muhammad, and Lei Jiao, "Capacity Driven SBS Deployment in Heterogeneous Cellular Networks: Outage probability and Rate coverage Analysis," *Transaction on Emerging Telecommunications Technologies*, vol. 31, no. 6, p. e3876, 2019. DOI: [10.1002/ett.3876](https://doi.org/10.1002/ett.3876)
- [J4]: Ziaul Haq Abbas, **Arif Ullah**, Ghulam Abbas, Fazal Muhammad, and Frank Yong Li, "Outage Probability Analysis of User-Centric SBS based HCNets Under Hybrid Rician/Rayleigh Fading," *IEEE Communication Letters*, vol. 24, no. 2, pp. 297-301, Dec. 2019. DOI: [10.1109/LCOMM.2019.2959578](https://doi.org/10.1109/LCOMM.2019.2959578)
- [J3]: **Arif Ullah**, Ziaul Haq Abbas, Ghulam Abbas, Fazal Muhammad, and Lei Jiao, "Performance Analysis of User-Centric SBS Deployment with Load Balancing in Heterogeneous Cellular Networks: A Thomas Cluster Process Approach," *Computer Networks*, vol. 170, pp. 107120, 2020. DOI: [10.1016/j.comnet.2020.107120](https://doi.org/10.1016/j.comnet.2020.107120)
- [J2]: **Arif Ullah**, Ziaul Haq Abbas, Fazal Muhammad, Ghulam Abbas, and Sunghwan Kim, "Uplink Performance Analysis of User-centric Small Cell Aided Dense HCNets with Uplink/Downlink Decoupling," *IEEE Access*, vol. 8, pp. 148460-148474, 2020. DOI: [10.1109/ACCESS.2020.3015915](https://doi.org/10.1109/ACCESS.2020.3015915)
- [J1]: Hammad Ahmad, Muhammad Mahmood Ali, **Arif Ullah**, Arbab Abdur Rahim, Husnul Maab, and Mahmood Khan, "An Ultra-Thin Beam Splitter Design Using a-Si:H Based on Phase Gradient Metasurfaces," *Journal of Nanoelectronics and Optoelectronics*, vol. 14, no. 9, pp. 1339-1343, Sep. 2019. DOI: [10.1166/jno.2019.2614](https://doi.org/10.1166/jno.2019.2614)

Manuscripts Submitted/in Preparation

- [S1]: **Arif Ullah**, Muhammad Ali Babar Abbasi and Hein Quoc Ngo, “Jamming Resilience in Fully Polarized Holographic MIMO Communications”, *Submitted to IEEE Transactions on Vehicular Technology*, 2025.

Conference Proceedings

- [C4]: **Arif Ullah**, Muhammad Ali Babar Abbasi, Okan Yurduseven, and Hein Quoc Ngo, “Holographic MIMO Communications with Orbital Angular Momentum and Polarization Multiplexing,” *2025 IEEE Globecom Workshops (GC Wkshps): Workshop on Integrated, Intelligent and Ubiquitous 6G Connectivity*, 08-12 Dec. 2025.
- [C3]: **Arif Ullah**, Muhammad Ali Babar Abbasi, and Hein Quoc Ngo, “Near-Field Holographic MIMO with Non-Paraxial Deployment in the Presence of Jamming,” *WS15 ESIT - IEEE International Conference on Communications (ICC) Montreal Canada*, 23-25 July. 2025. DOI: [10.36227/techrxiv.171174559.94756982/v1](https://doi.org/10.36227/techrxiv.171174559.94756982/v1)
- [C2]: Fawad, **Arif Ullah**, Iftikhar, and Wooyeol Choi, “RS-DeepNet: A Machine Learning Aided RSSI Fingerprinting for Precise Indoor Localization,” *The 2nd International Conference on Maritime IT Convergence*, Jeju, South Korea, 23-25 Aug. 2023. (**Outstanding paper award**) DOI: [10.36227/techrxiv.171174559.94756982/v1](https://doi.org/10.36227/techrxiv.171174559.94756982/v1)
- [C1]: **Arif Ullah**, Ziaul Haq Abbas, Ghulam Abbas, and Fazal Muhammad, “Analysis of Outage Probability and Rate Coverage in Heterogeneous Cellular Networks with joint uniform and clustered users,” *20th IEEE International Multi topic Conference (INMIC)*, Islamabad, Pakistan, 29-30 Nov. 2019. DOI: [10.1109/INMIC48123.2019.9022767](https://doi.org/10.1109/INMIC48123.2019.9022767)

Thesis

- [T1]: **Arif Ullah**. (2021). User-centric Small Cell Aided Future Cellular Networks: Sub-6GHz and Hybrid Millimeter Wave Communications [Doctoral dissertation, GIK Institute of Engineering Sciences and Technology]. Available online on HEC portal <http://pr.hec.gov.pk/jspui/handle/123456789/18249>
- [T2]: **Arif Ullah** (2016). Precise Estimation of Soft Output for Sphere Decoding based MIMO OFDM Receiver using Modified Likelihood Ascent Search Algorithm [MSc. dissertation, COMSATS University]. Institutional Repository at <https://....>



ACADEMIC PROJECTS

- **U.S.-Ireland R&D Partnership: INtegrated Sensing and Telecommunications for INtelligent Connection and Transmission (INSTINCT)**: Contributed to electromagnetic-informed channel modeling and signaling design part. (**NSF Proposal No: 2526414**), **Status**: Secured funding.
- **Modeling and Analysis of Intelligent V2E Network for 6G (MAIVNet-6G)**: Project submitted for MSCA fellowship. Unfortunately, this project couldn't secure funding in the Horizon Europe Program 2022, however, it received encouraging remarks, achieving an overall score of **78/100 (Excellence: 4.5/5, Impact: 3/5, Implementation: 3.7/5)** in evaluation.
- **Federated Deep Reinforcement Learning-based Resource Allocation for Flying Ad-hoc Networks**: This project was funded by the National Research Foundation (NRF), Republic of Korea.
- **User-centric Small Cells aided Multi-tier Networks**: This project focuses on the stochastic geometry modeling and performance evaluation of hotspot-aided user-centric small cell deployment in HCNet (2018–2021)
- **Beam Splitter Design using Metasurfaces**: This project focuses on the design of an ultra-thin beam splitter Using a-Si: H based on phase gradient metasurfaces in HFSS (Spring 2018)
- **Precise Estimation of Soft Output for Sphere Decoder**: In this project we precisely estimated the soft output for sphere decoding (SD) receiver in multi-antenna setup using low complexity modified likelihood ascent search algorithm (LAS) in MIMO OFDM system. (2015–2016)



COURSES UNDERTAKEN

Online Courses:

- (2021), “Introduction to Machine Learning”: offered by DUKE University online and Coursera

Graduate Courses:

- (2014): Stochastic Processes (EEE-611), Optimization Techqs. (EEE-712), Microwave Passive Devices & Circuits (ETN-611), EM Field Theory (ETN-610), Radio Engg. (ETN-616), Data Commun. & Nets. (ETN-671)
- (2015): Linear System Theory (ECI-665), Wireless Communication Techniques (ETN-644)
- (2017): Advance Algo. & Computational Techs. (CS-506), Organic Semiconductor & Devices (EE-633)
- (2018): Computational Methods for Engineers (ES-531), Instrumentation & Control Systems (EN-541), Cyber Security & IOT (CS-520), Electromagnetic Meta materials (EE-613)



SKILLS SUMMARY

- **Programming:** Python (basic), Matlab, L^AT_EX, C/C++, Assembly language, Mathematica
- **Softwares:** Simulink, Advance Design System (ADS), High Frequency Structured Simulation (HFSS), CST Microwave Studio, Pspice, Electronic Workbench, Inkscape, Linux
- **Languages:** English, Urdu, Pashto



HONORS, AWARDS, & MEMBERSHIPS

- **Postgraduate Fellowship:** Selected for Graduate Assistantship (GA4) during my PhD studies at GIK Institute of Engineering Sciences and Technology Pakistan, **2017 - 2021**
- **MS Scholarship:** Selected for Prime Minister Fee Reimbursement Scheme with full fee scholarship covering the tuition fee expenses during MS studies at COMSATS University, **2014 - 2016**
- **Member IEEE:** Member of Institute of Electrical and Electronics Engineers with [Membership #: 95038221](#)
- **Member PEC:** Member of Pakistan Engineering Council with [Membership #: ELECTRO/16479](#)



PRESENTATIONS/AWARDS/SEMINARS

- **P2:** Paper Presentation in the 2nd International Conference on Maritime IT Convergence, Jeju, South Korea, 2023
- **P1:** Paper Presentation in the 22nd IEEE International Multi Topic Conference held at National University of Computer and Emerging Sciences, Islamabad, Pakistan, 2019
- **C1:** Attended IEEE International Conference on Communications (ICC), Seoul, South Korea, 2022
- **W1:** Attended the 1st International Pak-Turk Workshop on Emerging Technologies in the Field of Science and Engineering held at GIK Institute of Engineering Sciences and Technology, Pakistan, 2018
- **W2:** Attended a hands on Workshop on Deep Intelligence, organized by Aerial Robotic Lab at GIK Institute of Engineering Sciences and Technology, Pakistan, 2021
- **S1:** Attended a seminar on “Writing a Good Research Paper” held at GIK Institute of Engineering Sciences and Technology, Pakistan, 2019



ORGANIZATIONS & COMMUNITY SERVICES

Journals/Conference Reviewer

- IEEE: Wireless Communications Letter, Transactions on Vehicular Technology, Transactions on Intelligent Transportation System, Communications Magazine, Transactions on Wireless Communication
- ELSEVIER: Digital Communications and Networks, Computer Networks
- MDPI: Drones, Sensors, Applied Sciences, Information
- [\(ICEAAI-2022\)](#): International Conference on Engineering Applications of Artificial Intelligence, Abha, KSA
- [\(ICCIT-2023\)](#): 3rd International Conference on Computing and Info. Techno, University of Tabuk, KSA

TPC

- [\(ICIESTR-2024\)](#): 1st International Conference on Innovative Eng. Sci. & Techno. Research, Muscat, Oman
- [\(INMIC-2023\)](#): 2nd International Conference on Maritime IT Convergence, Jeju South Korea
- [\(ICRAIS-2023\)](#): International Conference on Recent Advances in IT for Sustainable Development, Manipal India



REFERENCES

- **Prof. Wooyeol Choi** (Avisor): School of Computer Science and Engineering, College of Software, Chung-Ang University, Seoul 06974, Republic of Korea. Email: wchoi@cau.ac.kr, Phone:
- **Prof. Dr. Ziaul Haq Abbas** (Ph.D Advisor): Associate Professor, Faculty of Electrical Engineering, GIK Institute of Engineering Sciences and Technology Topi, 23640, Pakistan. Email: ziaul.h.abbas@giki.edu.pk
- **Prof. Dr. Shahid Khattak** (MS. Advisor): Professor and HOD Department of Electrical Engineering, COMSATS University 22060 Abbotabad, Islamabad, Pakistan. Email: skhattak@ciit.net.pk

